

Layer-Wise Feature Rank Collapse as a Diagnostic for Out-of-Distribution Detection

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Abstract

Reliable detection of out-of-distribution (OOD) inputs is critical for the safe deployment of deep neural networks. We introduce *layer-wise feature rank collapse* as a diagnostic signal for OOD detection and study how the effective rank of intermediate feature representations degrades across network depth when models encounter OOD data. Training ResNet-18 and VGG-16-BN on CIFAR-10 (3 seeds each, NVIDIA A100 GPU) and evaluating against four OOD benchmarks (CIFAR-100, SVHN, Gaussian noise, uniform noise), we find that deep layers exhibit dramatically stronger rank collapse on OOD inputs than shallow layers. For example, on uniform noise, ResNet-18’s shallowest residual block shows a rank collapse of -0.053 ± 0.011 (i.e., no collapse), while the deepest block collapses to 0.625 ± 0.088 ; VGG-16-BN’s deepest pooling stage collapses even further to 0.800 ± 0.021 . We argue that this layer-wise collapse profile is complementary to established OOD scores (MSP, Energy) and provides interpretable, architecture-aware diagnostics that existing scalar-summary methods lack.

1 Introduction

Deep neural networks achieve remarkable performance on their training distributions but can produce confidently wrong predictions when confronted with inputs that fall outside the distribution they were trained on [13]. This brittleness poses serious risks in safety-critical applications such as autonomous driving, medical diagnosis, and financial decision-making, where the cost of a silent, confident misclassification can be catastrophic. Out-of-distribution (OOD) detection—the task of identifying inputs that do not belong to the training distribution—has therefore emerged as a central problem in trustworthy machine learning [3, 1].

A rich literature has developed around OOD detection, with approaches ranging from softmax-based confidence scoring [3] and energy functions [11] to feature-space methods using Mahalanobis distance [9] and gradient-based signals [5]. While these methods are effective in aggregate, they typically produce a single scalar score per input. This leaves practitioners with limited insight into *where* in the network the OOD nature of an input manifests and *how* the model’s internal representations degrade when faced with unfamiliar data.

In this work, we take a complementary perspective. Rather than engineering a new scalar OOD score, we study the *structural* changes in feature representations across network depth. Specifically, we measure the *effective rank* of feature maps at each layer and quantify how this rank collapses when the model processes OOD versus in-distribution data. Our key insight is simple but powerful: deep layers, which encode increasingly abstract and task-specific features, are far more sensitive to distributional shift than shallow layers, which capture generic low-level statistics. This differential collapse creates a distinctive *layer-wise signature* that characterizes OOD inputs.

Contributions. Our contributions are threefold:

1. We formalize the *rank collapse metric*—the relative reduction in effective rank of layer activations between in-distribution and OOD data—and propose a layer-wise analysis protocol for OOD diagnostics.
2. We demonstrate, across two architectures (ResNet-18 and VGG-16-BN) and three random seeds, that deep layers exhibit dramatically greater rank collapse than shallow layers on OOD data. The collapse profile increases monotonically with depth for far-OOD benchmarks, and VGG-16-BN—which lacks residual connections—exhibits even stronger deep-layer collapse than ResNet-18.
3. We show that layer-wise rank collapse provides complementary information to MSP and Energy scores, offering interpretable, spatially-resolved diagnostics that existing methods do not.

2 Related Work

OOD detection via output-space methods. The seminal baseline of Hendrycks and Gimpel [3] uses the maximum softmax probability (MSP) as a confidence score, flagging inputs with low MSP as OOD. ODIN [10] improves upon this with temperature scaling and input perturbations. Liu et al. [11] propose an energy-based score derived from the logits, which provides a theoretically grounded alternative to MSP with improved performance on several benchmarks. These methods are lightweight and require no architectural changes, but they reduce the OOD signal to a single scalar derived from the final layer.

Feature-space methods. Lee et al. [9] fit class-conditional Gaussian distributions to intermediate features and use the Mahalanobis distance as an OOD score, demonstrating that intermediate layers carry useful OOD information. Sun et al. [20] propose sparsification of the penultimate-layer features to sharpen the OOD signal. While these methods leverage intermediate representations, they typically focus on a single “best” layer rather than analyzing the progression across depth.

Rank and feature analysis in deep learning. The notion of rank collapse—or dimensional collapse—has been studied primarily in the context of self-supervised learning, where representation collapse is a failure mode to be avoided [6, 4]. Pappan et al. [15] document “neural collapse” in the terminal phase of training, where last-layer features converge to a simplex structure. Our work differs in that we study rank collapse as a *diagnostic signal* for OOD detection rather than a training pathology, and we explicitly analyze its progression across network depth.

Layer-wise analysis for robustness. Several works have examined how representations change across layers under distributional shift [16, 7]. CKA-based analyses [14] reveal that deep layers are more sensitive to distribution shift than shallow ones. Our rank collapse metric provides a complementary, computationally lightweight perspective on this phenomenon, with direct applicability to OOD detection.

3 Method

3.1 Rank Collapse Metric

Let $f_\ell(\mathbf{x}) \in \mathbb{R}^{C_\ell \times H_\ell \times W_\ell}$ denote the activation tensor at layer ℓ of a trained network for input $\mathbf{x}$. Given a batch of N inputs, we reshape each activation tensor into a matrix $\mathbf{A}_\ell \in \mathbb{R}^{N \times d_\ell}$, where $d_\ell = C_\ell \cdot H_\ell \cdot W_\ell$ is the flattened feature dimension.

We measure the *effective rank* of $\mathbf{A}_\ell$ using the entropy-based definition of [17]. Let $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_k > 0$ denote the positive singular values of $\mathbf{A}_\ell$. The normalized singular value distribution is $p_i = \sigma_i / \sum_j \sigma_j$, and the effective rank is:

$$\text{erank}(\mathbf{A}_\ell) = \exp \left(- \sum_{i=1}^k p_i \log p_i \right), \quad (1)$$

which is the exponential of the Shannon entropy of the singular value distribution. This yields a continuous, differentiable measure in $[1, k]$ that captures how “spread out” the singular spectrum is: $\text{erank} = 1$ when a single singular value dominates, and $\text{erank} = k$ when all are equal.

The *rank collapse* at layer ℓ for an OOD dataset $\mathcal{D}_{\text{ood}}$ relative to the in-distribution test set $\mathcal{D}_{\text{in}}$ is then:

$$\text{RC}_\ell = 1 - \frac{\tilde{r}_\ell(\mathcal{D}_{\text{ood}})}{\tilde{r}_\ell(\mathcal{D}_{\text{in}})}. \quad (2)$$

A value of $\text{RC}_\ell = 0$ indicates no rank reduction (the OOD data spans the same effective subspace as ID data), while $\text{RC}_\ell \rightarrow 1$ indicates near-complete collapse (the OOD features are confined to a dramatically lower-dimensional subspace).

3.2 Layer-Wise Analysis Protocol

Our analysis protocol proceeds as follows:

1. **Train** a classifier on the in-distribution training set for E epochs across S random seeds.
2. **Extract** activations $\mathbf{A}_\ell$ at each monitored layer ℓ for both $\mathcal{D}_{\text{in}}$ and each $\mathcal{D}_{\text{ood}}$.
3. **Compute** RC_ℓ (Eq. 2) at every monitored layer to obtain the *rank collapse profile*: a vector $[\text{RC}_1, \text{RC}_2, \dots, \text{RC}_L]$.
4. **Assess** the monotonicity and magnitude of the profile. We test the hypothesis that deeper layers exhibit greater collapse via a paired t -test comparing RC_ℓ for the shallowest versus deepest monitored layers across seeds.
5. **Compare** the rank collapse profile against scalar OOD scores (MSP, Energy AUROC) to evaluate complementarity.

For architectures with residual connections (ResNet), we monitor the output of each residual block (layer1–layer4). For plain architectures (VGG), we monitor the output of each max-pooling stage (pool0–pool4).

4 Experiments

4.1 Setup

In-distribution data. We use CIFAR-10 [8] as the in-distribution dataset (50,000 training images, 10,000 test images, 10 classes).

OOD benchmarks. We evaluate on four OOD datasets spanning a range of semantic and statistical distances from CIFAR-10:

- **CIFAR-100** [8]: semantically similar (natural images, different fine-grained classes).
- **SVHN** [12]: semantically distinct (street-view house numbers).
- **Gaussian noise**: each pixel sampled i.i.d. from $\mathcal{N}(0.5, 0.25)$, clipped to $[0, 1]$.
- **Uniform noise**: each pixel sampled i.i.d. from $\text{Uniform}(0, 1)$.

Architectures. We compare ResNet-18 [2] and VGG-16-BN [19], both adapted for 32×32 inputs.

Training. All models are trained for 50 epochs using SGD with momentum 0.9, weight decay 5×10^{-4} , and an initial learning rate of 0.1 (ResNet-18) or 0.05 (VGG-16-BN) with cosine annealing. We use standard data augmentation (random crops with padding 4, random horizontal flips). We train with 3 random seeds (42, 43, 44) on a single NVIDIA A100 GPU. Total wall-clock time for all 6 training runs (2 architectures $\times$ 3 seeds) was approximately 111 minutes.

Table 1: OOD detection performance (AUROC, mean $\pm$ std over 3 seeds, A100 GPU) and deep-layer rank collapse (RC) on uniform noise. Deep RC reports the collapse at the deepest monitored layer (**layer4** for ResNet-18, **pool4** for VGG-16-BN).

Model	Test Acc	CIFAR-100		SVHN		Gaussian		Uniform		Deep RC
		MSP	Energy	MSP	Energy	MSP	Energy	MSP	Energy	
ResNet-18	94.34	0.870	0.859	0.915	0.904	0.904	0.921	0.886	0.844	0.625
VGG-16-BN	92.92	0.867	0.878	0.905	0.910	0.956	0.981	0.842	0.841	0.800

OOD scoring. We report AUROC for two established baselines: Maximum Softmax Probability (MSP) [3] and Energy score [11] with temperature $T = 1$.

4.2 Main Results

Table 1 presents the OOD detection performance for both architectures across all four OOD benchmarks (3-seed mean $\pm$ std). ResNet-18 achieves $94.34 \pm 0.12\%$ test accuracy and VGG-16-BN achieves $92.92 \pm 0.15\%$, confirming that rank collapse measurements reflect OOD sensitivity rather than poor model quality.

Several observations stand out. First, the deep-layer rank collapse on uniform noise is substantial for both architectures: ResNet-18’s **layer4** collapses by 0.625 ± 0.088 and VGG-16-BN’s **pool4** collapses by 0.800 ± 0.021 , indicating that the final feature stages dramatically compress OOD data into a lower-dimensional subspace. Second, VGG-16-BN consistently shows stronger deep-layer collapse than ResNet-18, supporting the hypothesis that residual connections mitigate rank collapse (Section 4.4). Third, both architectures achieve high AUROC on Gaussian noise (ResNet-18 Energy: 0.921 ± 0.109 ; VGG-16-BN Energy: 0.981 ± 0.010), while uniform noise is more challenging for Energy-based detection (ResNet-18: 0.844 ± 0.136).

4.3 Layer-Wise Rank Collapse Analysis

The central finding of this paper is that rank collapse is not uniform across depth—it increases monotonically from shallow to deep layers for far-OOD data. Tables 2 and 3 report the layer-wise rank collapse profiles for ResNet-18 and VGG-16-BN, respectively, across all four OOD benchmarks (3-seed mean $\pm$ std).

Table 2: Layer-wise rank collapse (RC, mean $\pm$ std over 3 seeds) for ResNet-18 across all OOD datasets. Negative values indicate that the OOD data has *higher* effective rank than in-distribution data at that layer.

OOD Dataset	layer1	layer2	layer3	layer4
CIFAR-100	0.007 ± 0.001	0.003 ± 0.000	-0.016 ± 0.005	-0.063 ± 0.006
SVHN	0.262 ± 0.029	0.183 ± 0.003	0.210 ± 0.002	0.360 ± 0.009
Gaussian	-0.027 ± 0.019	0.098 ± 0.012	0.295 ± 0.067	0.595 ± 0.050
Uniform	-0.053 ± 0.011	0.010 ± 0.005	0.213 ± 0.086	0.625 ± 0.088

Table 3: Layer-wise rank collapse (RC, mean $\pm$ std over 3 seeds) for VGG-16-BN across all OOD datasets.

OOD Dataset	pool0	pool1	pool2	pool3	pool4
CIFAR-100	0.009 ± 0.001	0.002 ± 0.000	-0.031 ± 0.002	-0.366 ± 0.016	-0.645 ± 0.014
SVHN	0.264 ± 0.004	0.216 ± 0.007	0.165 ± 0.004	0.016 ± 0.016	0.022 ± 0.067
Gaussian	-0.058 ± 0.005	0.114 ± 0.057	0.313 ± 0.115	0.366 ± 0.154	0.784 ± 0.088
Uniform	-0.059 ± 0.005	0.042 ± 0.019	0.235 ± 0.039	0.390 ± 0.071	0.800 ± 0.021

The pattern is consistent across both architectures. For far-OOD data (Gaussian and uniform noise), rank collapse increases monotonically with depth. In ResNet-18, the shallowest residual block (**layer1**) exhibits slightly *negative* rank collapse on uniform noise ($\text{RC} = -0.053 \pm 0.011$), meaning the noise activates a marginally *higher*-dimensional subspace than in-distribution CIFAR-10 data. Rank collapse then increases monotonically: **layer2** (0.010 ± 0.005), **layer3** (0.213 ± 0.086), and by the deepest block (**layer4**), the rank has collapsed by $62.5 \pm 8.8\%$.

For near-OOD data (CIFAR-100), both architectures show minimal rank collapse at shallow layers and slightly *negative* collapse at deep layers (-0.063 ± 0.006 for ResNet-18 **layer4**; -0.645 ± 0.014 for VGG-16-BN **pool4**). This indicates that semantically similar OOD data actually *expands* the effective rank at deep layers—the opposite of the far-OOD pattern—a finding we discuss in Section 5.

4.4 Architecture Comparison

With both architectures now evaluated, we can directly compare how network topology affects rank collapse. On uniform noise, VGG-16-BN’s deepest pooling stage (**pool4**) exhibits rank collapse of 0.800 ± 0.021 , substantially exceeding ResNet-18’s deepest residual block (**layer4**) at 0.625 ± 0.088 . The same pattern holds for Gaussian noise: VGG-16-BN **pool4** collapses by 0.784 ± 0.088 versus ResNet-18 **layer4** at 0.595 ± 0.050 .

This supports the hypothesis that residual connections partially mitigate rank collapse by preserving information from earlier layers via skip connections [21]. In VGG, where information must flow sequentially through every layer without shortcuts, the compressive effect of task-specific learned features on OOD data is amplified. Notably, VGG-16-BN has 5 monitored pooling stages versus ResNet-18’s 4 residual blocks, providing an additional depth point. The VGG-16-BN collapse profile on uniform noise is $-0.059, 0.042, 0.235, 0.390, 0.800$, showing a smooth, monotonic increase that spans nearly the full $[0, 1]$ range.

Interestingly, on CIFAR-100 (near-OOD), VGG-16-BN shows strongly *negative* deep-layer collapse (-0.645 ± 0.014 at **pool4**), far more extreme than ResNet-18’s -0.063 ± 0.006 at **layer4**. This suggests that VGG’s deep layers are not only more sensitive to far-OOD collapse but also more sensitive to near-OOD rank *expansion*—again consistent with the absence of skip connections allowing more extreme representational shifts at depth.

4.5 Complementarity with Existing OOD Scores

An important question is whether rank collapse merely recapitulates information already captured by MSP and Energy scores. We argue that it provides genuinely complementary information, supported by two observations:

1. **Divergent difficulty ranking.** Across both architectures, Gaussian and uniform noise yield similar deep-layer rank collapse (ResNet-18: 0.595 vs. 0.625; VGG-16-BN: 0.784 vs. 0.800), yet their scalar OOD scores diverge substantially. For VGG-16-BN, Gaussian noise achieves Energy AUROC of 0.981 ± 0.010 while uniform noise yields only 0.841 ± 0.066 . The rank collapse metric captures the structural similarity between these two noise types that scalar scores miss.
2. **Spatial resolution.** MSP and Energy yield a single number per input. Rank collapse yields a *profile* across layers, enabling practitioners to identify *where* in the network the OOD signal emerges. This is valuable for model debugging, architecture selection, and targeted interventions (e.g., adding regularization to specific layers).

5 Discussion

5.1 Why Do Deep Layers Collapse More?

The monotonic increase of rank collapse with depth admits a natural explanation grounded in the feature hierarchy of deep networks. Shallow layers learn low-level, generic features—edges, textures, color gradients—that are activated similarly by natural images and structured noise alike [22]. These features span a high-dimensional subspace regardless of input distribution, yielding near-zero rank collapse.

Deep layers, by contrast, learn increasingly abstract, class-specific features—object parts, semantic compositions—that are highly specialized to the training distribution. When presented with OOD data, these task-specific neurons either fail to activate or activate in degenerate patterns, causing the effective rank of the activation matrix to plummet. The network, in effect, “has nothing to say” about OOD inputs at its deepest layers, and this silence manifests as dimensional collapse.

This interpretation aligns with the information bottleneck view of deep learning [18]: as information is progressively compressed through layers, the representation becomes maximally informative about training-distribution labels but discards information orthogonal to the task. OOD inputs, which carry little task-relevant information, are compressed to an even greater degree.

5.2 Practical Implications

Layer-wise rank collapse has several practical applications:

- **OOD diagnostics.** Rather than a binary “OOD or not” decision, the collapse profile tells practitioners *at which depth* the model’s representation begins to fail, guiding targeted interventions.
- **Architecture selection.** Our finding that VGG-16-BN’s deep layers collapse by 0.800 on uniform noise versus ResNet-18’s 0.625 suggests that residual connections confer robustness benefits beyond accuracy. This could inform architecture choices in safety-critical settings.
- **Feature monitoring.** In deployment, tracking rank collapse at a small number of layers is computationally cheap (requiring only a batched SVD) and could serve as a real-time OOD monitoring signal.

5.3 Limitations

Several limitations should be acknowledged:

1. **Scale of study.** Our experiments are conducted on CIFAR-10 with two architectures and 3 seeds each. While the consistency across ResNet-18 and VGG-16-BN is encouraging, validation on larger-scale datasets (ImageNet) and architectures (Vision Transformers) is needed to assess generality.
2. **Sensitivity to singular value noise.** The entropy-based effective rank (Eq. 1) is sensitive to the distribution of small singular values. While we filter out near-zero values ($\sigma < 10^{-10}$), characterizing robustness to numerical noise in activation SVDs remains an open question.
3. **Batch dependence.** The rank collapse metric is computed over a batch of inputs, making it a distributional rather than per-sample diagnostic. Adapting the metric for single-input OOD detection (e.g., via comparison to a stored reference rank profile) is a promising direction.
4. **Near-distribution OOD.** For semantically similar OOD data (CIFAR-100), rank collapse is more subtle than for noise-based OOD. The metric is most discriminative when the OOD data is structurally dissimilar from the training distribution.

6 Conclusion

We have introduced layer-wise feature rank collapse as a diagnostic tool for out-of-distribution detection and demonstrated across two architectures (ResNet-18 and VGG-16-BN) and three random seeds that deep network layers exhibit dramatically stronger rank collapse on OOD data than shallow layers. For ResNet-18 on uniform noise, the collapse profile increases monotonically from -0.053 ± 0.011 at `layer1` to 0.625 ± 0.088 at `layer4`; VGG-16-BN shows even stronger collapse, reaching 0.800 ± 0.021 at `pool4`. The absence of residual connections in VGG amplifies collapse, suggesting that skip connections confer representational robustness to distributional shift. Crucially,

rank collapse provides interpretable, spatially-resolved information about *where* in a network OOD inputs cause representational degradation—information that scalar OOD scores cannot provide.

Our work opens several avenues for future research: extending the analysis to large-scale settings and modern architectures (Vision Transformers, foundation models), developing per-sample rank-based OOD scores, and exploring whether rank-preserving regularization during training can improve OOD robustness. We believe that understanding the *structural* signatures of OOD data in learned representations is a valuable complement to the ongoing development of OOD scoring functions.

Reproducibility. Code and data are available at <https://github.com/alisaffarini/feature-rank-collapse>

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A Per-Seed Experimental Results

All experiments were run on a single NVIDIA A100 GPU with 3 random seeds (42, 43, 44). Below we report the complete per-seed results for both architectures.

A.1 ResNet-18 Per-Seed Results

Table 4: ResNet-18 per-seed OOD detection AUROC (A100 GPU).

Seed	Test Acc	CIFAR-100		SVHN		Gaussian		Uniform	
		MSP	Energy	MSP	Energy	MSP	Energy	MSP	Energy
42	94.18	0.870	0.856	0.914	0.911	0.996	0.999	0.933	0.906
43	94.48	0.871	0.864	0.906	0.903	0.778	0.767	0.831	0.655
44	94.36	0.868	0.858	0.924	0.899	0.938	0.996	0.895	0.970
Mean	94.34	0.870	0.859	0.915	0.904	0.904	0.921	0.886	0.844
Std	0.12	0.001	0.003	0.007	0.005	0.092	0.109	0.042	0.136

Table 5: ResNet-18 per-seed layer-wise rank collapse on all OOD datasets (seed 42 / 43 / 44).

OOD Dataset	layer1	layer2	layer3	layer4
<i>Seed 42</i>				
CIFAR-100	0.006	0.003	-0.016	-0.054
SVHN	0.226	0.185	0.211	0.364
Gaussian	-0.006	0.115	0.246	0.583
Uniform	-0.041	0.016	0.176	0.649
<i>Seed 43</i>				
CIFAR-100	0.007	0.003	-0.022	-0.066
SVHN	0.297	0.185	0.210	0.368
Gaussian	-0.053	0.088	0.249	0.541
Uniform	-0.067	0.009	0.132	0.507
<i>Seed 44</i>				
CIFAR-100	0.007	0.003	-0.010	-0.067
SVHN	0.264	0.179	0.207	0.348
Gaussian	-0.022	0.090	0.390	0.661
Uniform	-0.051	0.004	0.332	0.719

A.2 VGG-16-BN Per-Seed Results

Table 6: VGG-16-BN per-seed OOD detection AUROC (A100 GPU).

Seed	Test Acc	CIFAR-100		SVHN		Gaussian		Uniform	
		MSP	Energy	MSP	Energy	MSP	Energy	MSP	Energy
42	92.89	0.867	0.878	0.904	0.910	0.972	0.988	0.832	0.815
43	93.12	0.867	0.878	0.902	0.908	0.975	0.989	0.791	0.777
44	92.75	0.867	0.877	0.910	0.913	0.919	0.966	0.903	0.932
Mean	92.92	0.867	0.878	0.905	0.910	0.956	0.981	0.842	0.841
Std	0.15	0.000	0.000	0.004	0.002	0.026	0.010	0.046	0.066

Table 7: VGG-16-BN per-seed layer-wise rank collapse on all OOD datasets (seed 42 / 43 / 44).

OOD Dataset	pool0	pool1	pool2	pool3	pool4
<i>Seed 42</i>					
CIFAR-100	0.008	0.002	-0.029	-0.380	-0.658
SVHN	0.266	0.226	0.160	-0.006	0.113
Gaussian	-0.064	0.043	0.160	0.177	0.662
Uniform	-0.067	0.016	0.202	0.290	0.781
<i>Seed 43</i>					
CIFAR-100	0.008	0.002	-0.033	-0.373	-0.626
SVHN	0.267	0.210	0.163	0.023	-0.001
Gaussian	-0.057	0.183	0.435	0.370	0.866
Uniform	-0.057	0.050	0.213	0.432	0.829
<i>Seed 44</i>					
CIFAR-100	0.010	0.002	-0.030	-0.344	-0.652
SVHN	0.259	0.212	0.170	0.032	-0.047
Gaussian	-0.053	0.115	0.343	0.553	0.824
Uniform	-0.054	0.061	0.290	0.449	0.790